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Improving Cement Log Response Through Sandblasting

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Abstract

Achieving reliable cement-to-casing bonding remains a persistent challenge, particularly in wells where microannulus is observed in cement evaluation logs. In the Caspian region, repeated occurrences of microannulus in 7 in liners prompted a focused investigation into surface preparation techniques and their effect on cement bonding. Despite optimizing cement design and mud removal improvements, dry microannulus – approximately 10 microns wide – continued to appear, despite prior optimization efforts.

To identify the root cause, a field trial was conducted comparing factory-coated and sandblasted liners placed in sequence within the same well. This allowed direct evaluation of bonding performance under identical downhole conditions. Results from ultrasonic logs demonstrated a clear improvement in bonding across sandblasted sections, while coated joints consistently showed dry microannulus. These findings suggest that liner surface condition plays a critical role in bond quality. The combination of sandblasting with other best practices and job design measures proved effective in mitigating microannulus and enhancing zonal isolation.

Introduction

Zonal isolation is a critical component throughout the life of a well. Improper isolation can lead to production loss, water cut, or gas leakage, particularly in mature reservoirs. Achieving effective zonal isolation across multiple pressure zones in depleted fields presents a major challenge. In the Caspian region, a major operator has faced recurring occurrences of microannulus in cement-to-casing bonds, as identified in ultrasonic evaluation logs. The issue is mostly noticed as dry microannuli – tiny gas filled gaps between the cement and casing – that weaken ultrasonic log responses and may compromise long-term zonal isolation. This phenomenon increases the risk of inter-zonal communication and jeopardizes well integrity.

To address the issue, several best practices were implemented such as reducing post-job downhole stresses, optimizing cement mechanical properties via flexible slurries, and improving mud removal. Despite these efforts, microannuli were still observed, prompting a re-evaluation of the root causes and potential mitigation methods. The objective of this paper is to address this challenge and demonstrate improvement of cement-to-casing bonding, eliminating microannulus across multi-pressure zones in depleted reservoirs through an unconventional approach.

Understanding Microannulus in Cement Evaluation Log

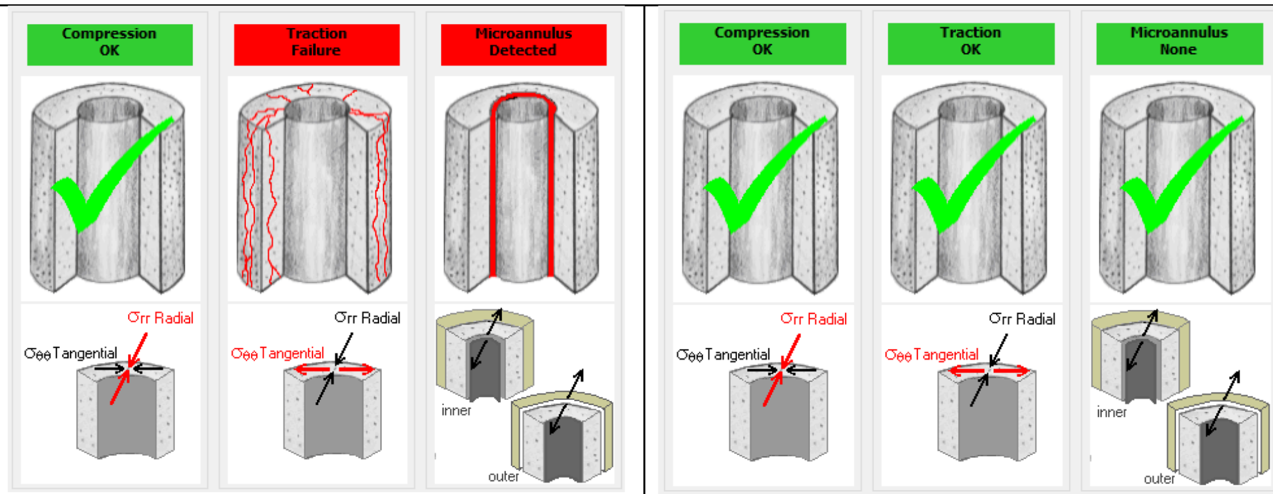
Microannulus is defined as a very thin gap, typically less than a few hundred micrometers, that forms between the set cement sheath and casing or formation. It is detected using ultrasonic logs and can act as a pathway for gas migration that compromises zonal isolation.

Downhole Stresses

Microannulus can occur at any stage of a well's life due to changes in wellbore conditions. A common cause is casing contraction resulting from pressure reduction after cement setting, often triggered by replacing the casing fluid with a lower-density fluid. Cooling of the wellbore can have a similar effect. A typical case involves shutting in the casing after cement placement. Heat generated by cement hydration expands the casing and warms trapped fluids, increasing expansion. Once the cement sets and the casing is opened, internal pressure drops and hydration heat dissipates, allowing the casing to contract to its original diameter.

Accordingly, in cased-hole environments, operations conducted after cement setting such as pressure tests, inflow testing, and well fluid displacement to lighter densities, impose additional stresses that increase the risk of microannulus formation. Fluctuations in downhole pressure and temperature may disturb the cement sheath, causing partial debonding at the casing interface. To mitigate this risk, it is essential to limit post-job activities that could compromise the cement, as such disturbances may trigger or worsen microannulus development.

Additional precaution to minimize risk of microannulus occurrence is stress resistant flexible cement systems with improved mechanical properties. As documented in multiple SPE papers such as SPE-92361-MS, SPE-119960-MS, and SPE-223654-MS flexible cement systems with low Young's modulus and expansion properties demonstrate improved resistance to stress conditions compared to conventional cement systems. This concept is further supported by stress analysis software simulations (figure 1) and field observations, where minimal microannulus was observed in wells where flexible slurry systems with improved mechanical properties were used. Mechanical properties analysis included Young's modulus, Poisson's ratio, and compressive strength of cement and formation. In below software simulation (figure 1) demonstrated comparison of conventional and flexible cement systems to stress responses under thermal and pressure changes. It demonstrates that flexible cement can prevent shear and tensile failures, thereby reducing microannulus formation.



Compression:

Cement will not fail in compression. Maximum Mohr-Coulomb compressive stress encountered after 4.2 min at inner casing-cement boundary in inner cement (3.5 in from well axis).

Maximum Mohr-Coulomb Stress: 587 psi.

Compressive strength of cement: 5000 psi.

Traction:

Cement will fail in traction initially after 3 min at inner casing-cement boundary in inner cement (3.5 in from well axis). At the time of failure well pressure change was 5000 psi and well temperature was 50 degC. Maximum tangential tensile stress: 171 psi. Tensile strength of cement: 150 psi.

Microannulus:

Max Microannulus width 48.1 µm. At the time of failure well pressure change was 500 psi and well temperature was 50 degC.

Start Time, min	End Time, min	Type	Start Pressure, psi	End Pressure, psi
0	0	Instant	0	500
0	3	Constant	500	500
3	3	Instant	500	5000
3	13	Constant	5000	5000
13	13	Constant	0	0

Compression and Traction:

The cement sheath is stress free.

Compression and Traction:

The cement sheath is stress free.

Microannulus:

No Microannulus detected

Start Time, min	End Time, min	Type	Start Pressure, psi	End Pressure, psi
0	0	Instant	0	500
0	3	Constant	500	500
3	3	Instant	500	5000
3	13	Constant	5000	5000
13	13	Constant	0	0

Figure 1—Cement Stress Analysis This figure 1(SPE-198580-MS) compares stress responses for conventional and flexible cement systems under thermal and pressure changes. It demonstrates that flexible cement can prevent shear and tensile failures, thereby reducing microannulus formation.

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Cement Shrinkage

Another contributing factor to microanulus formation is cement sheath volume loss from chemical and bulk shrinkage. Bulk shrinkage happens while cement is in liquid state while chemical shrinkage continues even after cement is set. This chemical shrinkage is driven by continuation of hydration process within set cement matrix. Total cement shrinkage can reach up to 6%. This was documented by Parcevaux and Sault (1984). Research indicates that the use of expanding agents in cement slurries can effectively compensate for chemical shrinkage. A controlled, slight expansion of the cement beyond its initial set volume, driven by crystal growth, offsets volume loss and strengthens both the cement-to-casing and cement-to-formation bonds. However, the application of expanding agents must be approached with caution, as several risks need to be considered. Excessive expansion can cause cracking of the cement, leading to loss of zonal isolation. In soft formations, the expansion force may push the cement outward, creating microannuli at the inner cement-to-casing interface. SPE-198630 (Bugrayev et al.) provides a clear demonstration of the benefits of expansion additives, showing improved cement bond quality over time, as confirmed by repeat logging performed after a defined period of well operation. An additional example of the effectiveness of expanding agents was observed in a well in 2023, where an ultrasonic log was repeated 23 days after the initial run. The repeat cement bond log over the same interval showed a clear improvement compared to the first results. These findings demonstrate that expanding agents can enhance cement bond quality over time, as evidenced by subsequent log responses (Figure 2).

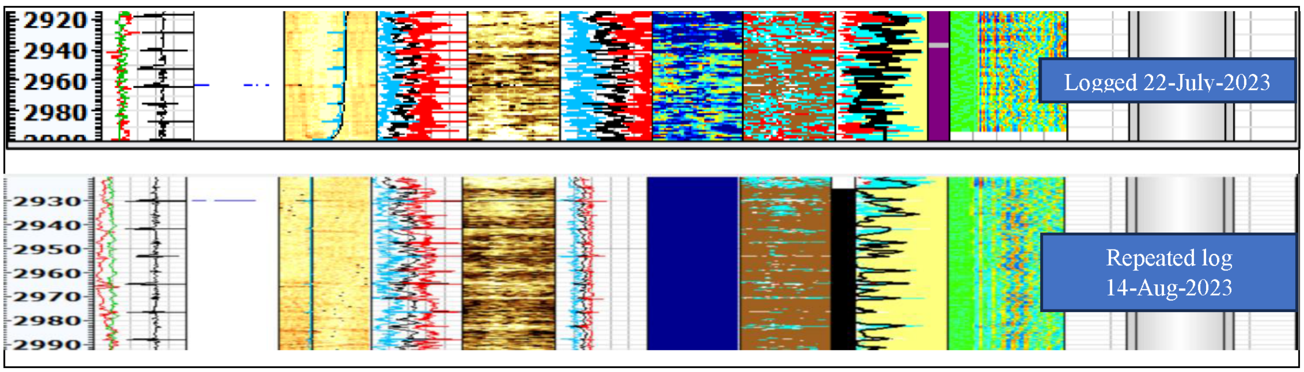

Figure 2—Slurry Expansion. An ultrasonic log was recorded on July 22, three days after the job. A repeat log conducted 23 days later showed a clear improvement in bond quality.

Size of microannulus can range in size depending on formation stress and casing deformation. In general, microannuli can be up to 100 microns wide. In our cases, consistently ultrasonic log recorded around 10 microns of dry microannulus.

Mud Removal and Spacer Optimization

Proper mud removal is essential to achieve good cement bonding. Presence of residual mud layers on the surface formation or casing surface act as a barrier to bonding.

Several factors contribute to poor mud removal: uncertain open hole geometry, insufficient liner stand-off, absence rheology and density hierarchies, inability to rotate the liner, and the absence of a mechanical separator ahead of slurry (dart) in liner cementing. In the cementing industry, the impact of these factors is well recognized, and the mitigation measures have been successfully applied. In studied location, measures for the above mentioned factors were implemented. Particular attention was given to liner rotation. Liner rotation plays a significant role in radial fluid flow around liner. In addition, spacer volume was optimized, and abrasive materials were added to the spacer. The use of abrasive fiber material was an additional assistant to remove mud from the surface of casing or formation by their additional mechanical scrubbing capability and absorption of mud through hydrophobic interaction.

By implementing the above mentioned practices, cement bonding was improved. However, despite applying best practices and executing jobs as planned, cement evaluation logs still indicated weak cement-to-casing adhesion in certain intervals. In particular, ultrasonic logs revealed the presence of a dry microannulus – interpreted as a gas-filled gap of approximately 10 microns between the cement and casing (Figure 3). This example illustrates the typical log response associated with dry microannulus. The presence of air or gas-filled microgaps can distort ultrasonic log interpretation, making the bond appear poor even when circumferential coverage exists. Although such gaps are not continuous fluid channels, they can still permit gas migration over time and compromise long-term zonal isolation. These observations prompted further investigation into the possible root causes, forming the basis for the following hypothesis.

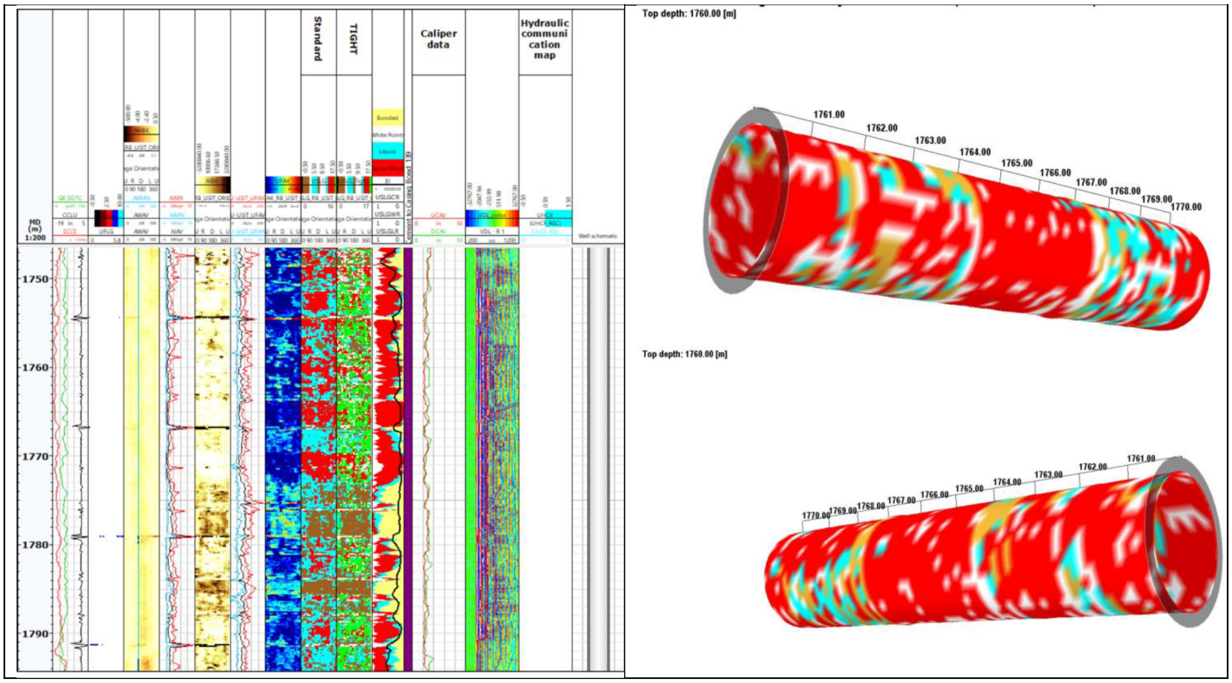

Figure 3—Dry Microannulus. This log example highlights the typical response associated with dry microannulus. The presence of gas or air-filled micro gaps affects ultrasonic log interpretation, making it appear as a poor bond even when coverage exists. These gaps, though not fluid paths, may still allow gas migration and compromise long-term isolation.

Hypothesis: Coating as Root Cause

Despite implementing industry best practices—such as improved mud removal, optimized spacer design, and the use of flexible cement systems—ultrasonic logs consistently indicated the presence of dry microannuli, typically around 10 microns wide. Importantly, these responses appeared even when all job parameters were executed according to design, and no post-job downhole stresses were applied. This repeated observation suggested that conventional explanations such as stress-induced debonding or job design limitations were unlikely to be the sole contributors. Therefore, attention turned toward alternative factors that could systematically influence cement-to-casing bonding.

One aspect identified for further investigation was the surface condition of casing and liners. In the studied field casing and liners were coated with a thin corrosion-protective layer, typically varnish or epoxy, which is commonly applied for storage and handling as permitted by API 5CT. This smooth coating typically 15 to 50 microns thick raised concern regarding potential impact on cement bonding. The reason for such concern was ultrasonic log observations, where dry microannulus frequently appeared with a gap of around 10 microns, which is similar to the minimum coating thickness size. This consistent observation further reinforced the hypothesis that the coating itself may be a contributing factor to poor cement-to-casing bonding. To verify this hypothesis, a field trial was conducted.

Sandblasting Application and Results

To validate this hypothesis, it was necessary to remove the factory applied coating. This was achieved through sandblasting. Sandblasting (abrasive blasting) is a widely used surface preparation method across multiple industries, including oilfield. It removes contaminants like mill varnish, rust, and creates a roughened surface to improve mechanical interlock. In this field trial, dry sandblasting was applied using copper slag abrasive grit (0.5 mm to 2.5 mm) to remove the coating and roughen the liner surface.

Case 1

The following 7in liner cementing job was performed with conventional design as previous jobs:

- 200 bbl of 12.7 ppg low-viscosity mud
- 100 bbl of 14.2 ppg Spacer
- 156.5 bbl of 15.8 ppg gas-tight cement slurry with expansion additive

Since cement bonding depends on factors such as formation type, centralization, and loss zones, it was decided to run three coated and three sandblasted liner/casing joints consecutively, followed by the rest. This approach allowed us to eliminate the effect of external influences on cement bonding – such as formation properties, centralization, and formation fluid or gas – during evaluation. By placing both types of joints in sequence within the same intervals, we ensured that all were subjected to identical downhole conditions, allowing for a direct and reliable comparison.

The cementing job was executed as planned with no losses. An ultrasonic cement bond log was performed 70 hours after the job (figure 4). The results revealed a clear contrast: dry microannulus was detected in intervals with coated liners, whereas sandblasted sections exhibited excellent bond responses. This outcome served as a strong demonstration, confirming the hypothesis that the liner coating was the primary cause of the dry microannulus observed in the logs.

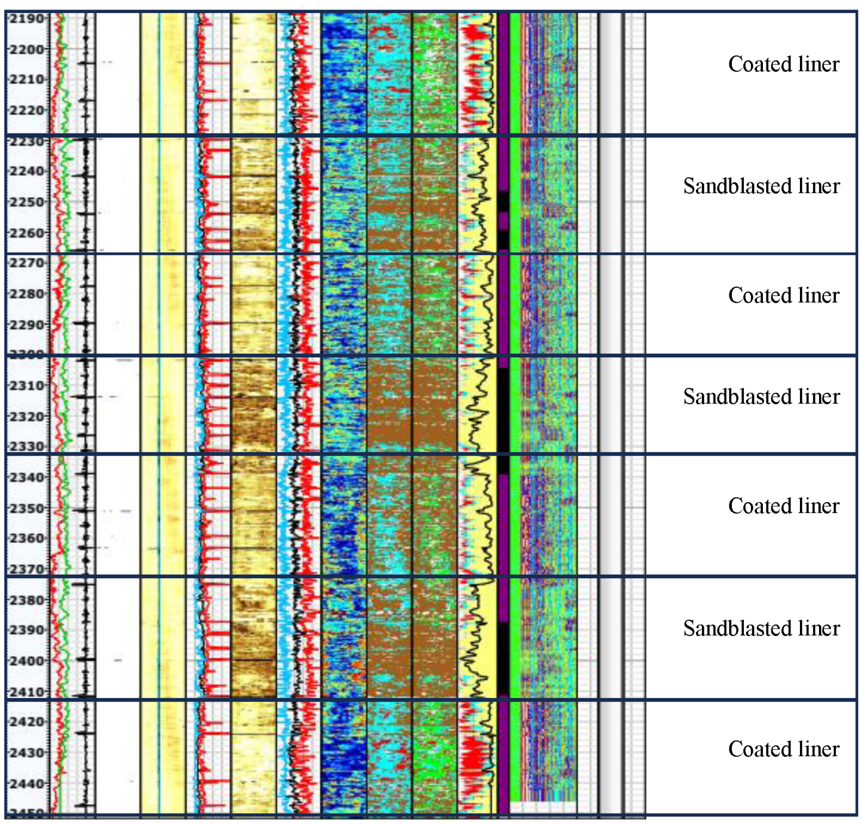

Figure 4—Ultrasonic log Case #1. Sandblasted liners show superior bonding, while coated sections display signs of dry microannulus.

Case 2

Another 7-in. liner was cemented following the established field practices and design. An optimized spacer volume with low-viscosity properties was used, while density and rheology hierarchies were properly maintained. Liner stand-off was within the recommended limits (>75%), and liner rotation was applied to improve mud removal. To minimize cement shrinkage, an expanding agent was included in the slurry design. The operation was executed successfully with no incidents or deviations.

In this case, sandblasted liners were run across the 1990 – 2420 m interval, while coated liners were placed above and below this section. Ultrasonic log results again demonstrated a clear difference. The intervals containing coated liners showed evidence of dry microannulus, whereas the sandblasted liner interval exhibited a distinctly improved bond response (Figure 5).

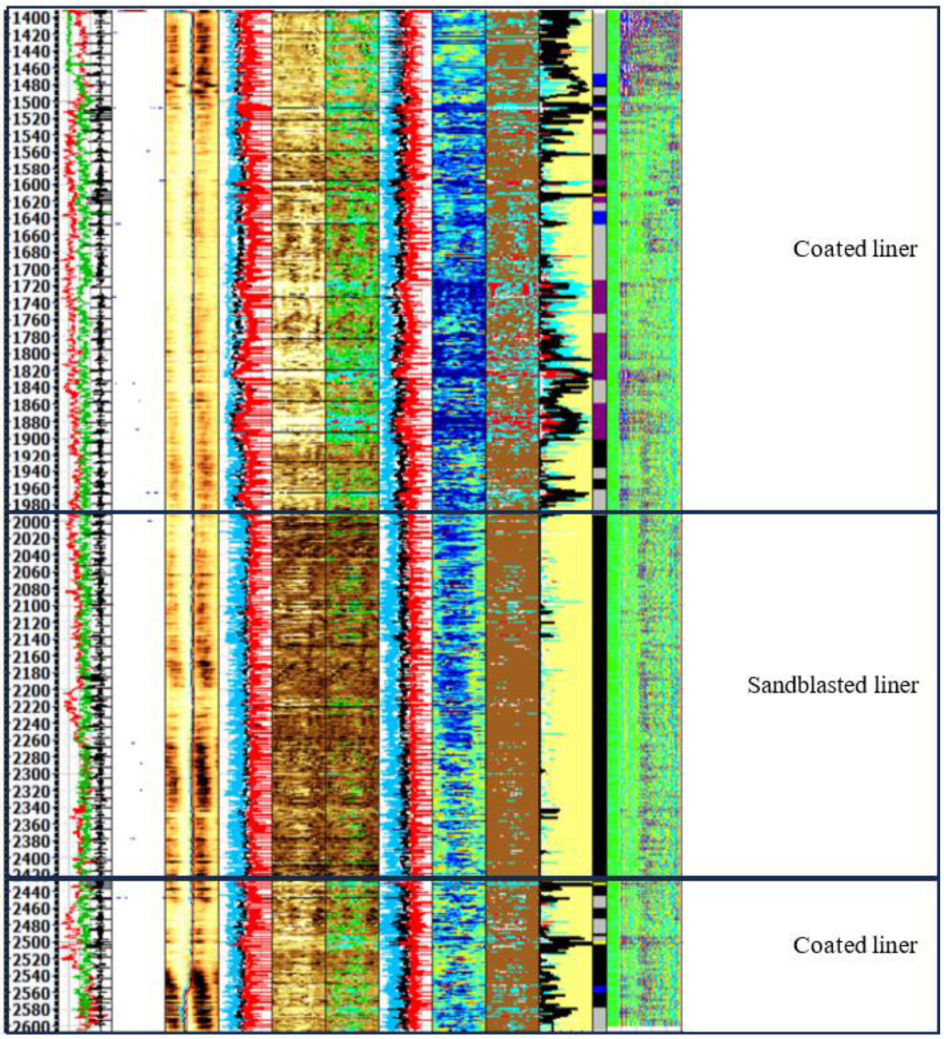

Figure 5—Ultrasonic log Case #2. Clearly seen effect of sandblasting liners.

This case provided further confirmation that liner coating can be the cause of dry microannulus observed in log results, which may lead to a misleading interpretation of poor bonding even when the liner is radially cemented.

Case 3

A third application was carried out to further validate the influence of liner surface condition on cement bonding. The cementing design followed all previously established best practices, including optimized spacer with abrasive fibers, density and rheology hierarchy, liner stand-off >75%, liner rotation, and the use of an expanding agent to minimize shrinkage. The cementing operation was executed without losses or deviations from agreed job design.

Ultrasonic cement evaluation logs (Figure 6) showed that the coated liner interval in this case did not exhibit severe bonding problems, with responses indicating generally good cement placement. However, a clear differentiation was observed when compared with the adjacent sandblasted liner intervals. The sandblasted sections demonstrated distinctly stronger and more uniform bond responses, with no signs of dry microannulus. This side-by-side contrast confirmed that while coated liners may sometimes achieve moderate bonding, sandblasting provides a consistently superior surface condition, resulting in improved log quality and stronger cement-to-casing adhesion.

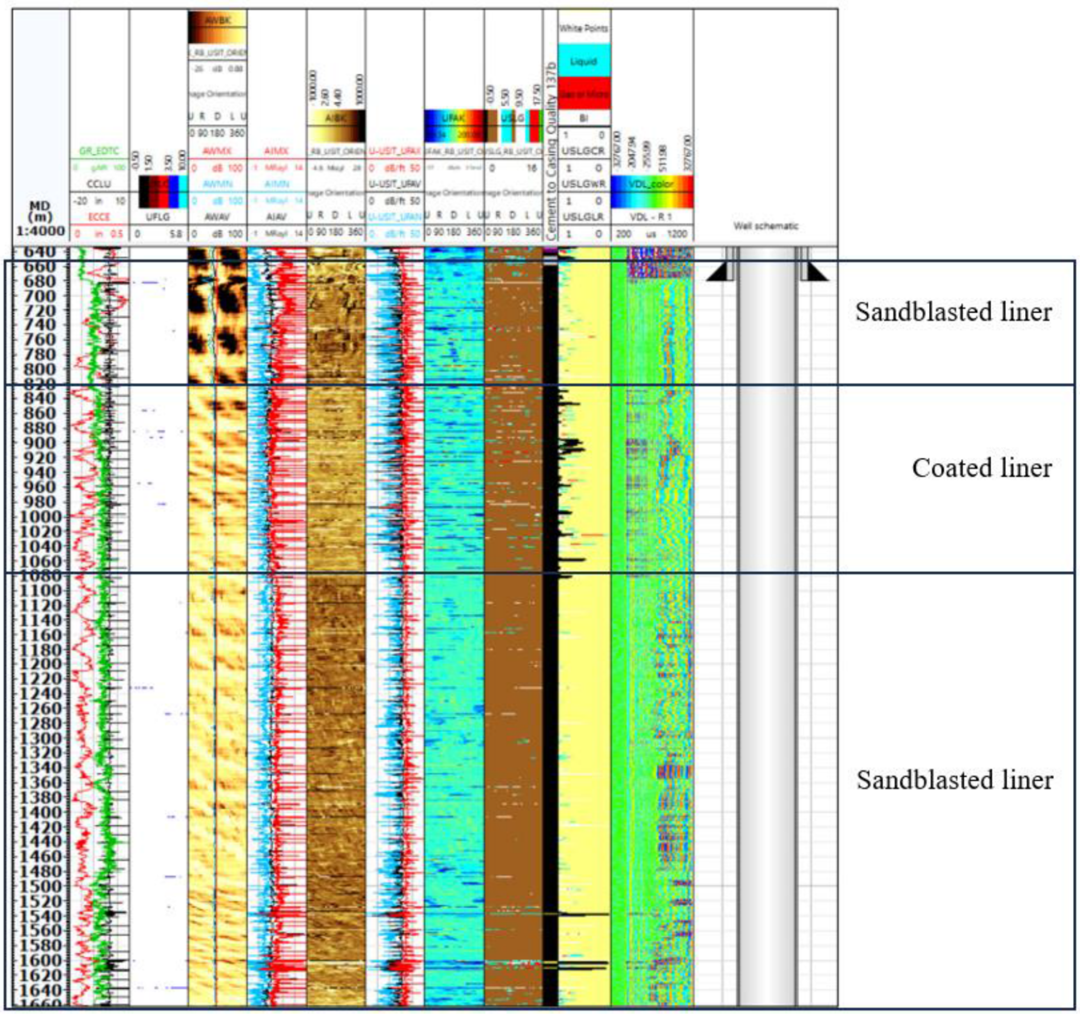

Figure 6—Figure 6 – Ultrasonic cement evaluation log comparison of sandblasted and coated liners.

This case further reinforced the conclusion that sandblasting not only eliminates the risk of coating-related microannulus but also enhances bond performance even under conditions where coated liners appear adequate.

Results and Interpretation

Ultrasonic logs revealed a clear contrast: dry microannulus was consistently detected in intervals with coated liners, whereas sandblasted sections showed an excellent bond response. Repeated runs confirmed the consistency of this outcome, demonstrating that the surface condition of the liner or casing was a critical factor. In both cases shown above, the liner was effectively cemented; however, due to the presence of coating, the log response appeared as microannulus. This finding highlighted the log data may indicate poor bonding despite adequate radial cement coverage across the liner or casing.

Integration into Routine Practice

Following the initial trials, sandblasting became a routine part of liner preparation. It was planned in advance and primarily applied in reservoir and perforation sections, where cement integrity is most critical. The process was smoothly integrated into normal workflows without adding extra time or requiring major operational changes. Ultrasonic logs consistently confirmed improved bond quality in sandblasted sections.

Selective application of sandblasting was necessary to balance technical benefits with long-term considerations. Factory-applied coatings help protect liners from corrosion during extended storage and removing them entirely could increase the risk of corrosion if liners are exposed or if losses prevent complete cement coverage. For this reason, sandblasting was selectively applied only to critical intervals, ensuring both reliable cement bonding in key zones and continued corrosion protection in less sensitive sections.

Conclusion

This study demonstrated that factory-applied liner coatings can contribute to the formation of dry microannuli observed in ultrasonic cement bond logs, even when best practices and advanced cementing systems are applied. Field trials showed that the smooth coating layer reduced cement-to-casing adhesion, resulting in misleading log responses and raising concerns for long-term zonal isolation.

Sandblasting proved effective in removing coatings and roughening the liner surface, enabling stronger cement bonding. Comparative results consistently confirmed improved log responses in sandblasted intervals, while coated sections continued to show microannuli. Repeated success across multiple wells validated the reliability of this approach.

Today, sandblasting is applied as a routine practice in critical well sections such as reservoirs and perforation zones, while coatings are maintained in less critical areas to preserve corrosion protection. This selective application balances operational reliability with long-term material integrity. Overall, the findings highlight the critical role of liner surface preparation in cementing performance and demonstrate that sandblasting in key zones significantly improves bonding, strengthens log evaluation confidence, and supports long-term well integrity.

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